

CURRICULUM VITAE

ANDREI PROKHOROV

CURRENT POSITION

2026–present Postdoctoral Fellow at the Baylor University

RESEARCH INTERESTS

Applications of Riemann-Hilbert problems in probability models and differential equations.

REFEREED PUBLICATIONS

- H. Pan and A. Prokhorov. “Asymptotic properties of special function solutions of the Painlevé III equation for fixed parameters”. In: *Stud. Appl. Math.* 154.4 (2025), Paper No. e70051, 32. ISSN: 0022-2526,1467-9590. MR: [4897093](#). arXiv: [2407.04852](#) [[math.CA](#)]
- H. Desiraju, A. Its, and A. Prokhorov. “Nonlinear steepest descent on a torus: a case study of the Landau-Lifshitz equation”. In: *Nonlinearity* 38.4 (2025), Paper No. 045023, 51. ISSN: 0951-7715,1361-6544. MR: [4890867](#). arXiv: [2405.17662](#) [[math.AP](#)]
- A. Barhoumi, O. Lisovyy, P. D. Miller, and A. Prokhorov. “Painlevé-III monodromy maps under the $D_6 \rightarrow D_8$ confluence and applications to the large-parameter asymptotics of rational solutions”. In: *SIGMA Symmetry Integrability Geom. Methods Appl.* 20 (2024), Paper No. 019, 77. ISSN: 1815-0659. MR: [4714353](#). arXiv: [2307.11217](#) [[math.CA](#)]
- J. Baik, A. Prokhorov, and G. L. F. Silva. “Differential equations for the KPZ and periodic KPZ fixed points”. In: *Comm. Math. Phys.* 401.2 (2023), pp. 1753–1806. ISSN: 0010-3616,1432-0916. MR: [4610285](#). arXiv: [2208.11638](#) [[math.PR](#)]
- E. C. Bailey, S. Bettin, G. Blower, J. B. Conrey, A. Prokhorov, M. O. Rubinstein, and N. C. Snaith. “Mixed moments of characteristic polynomials of random unitary matrices”. In: *J. Math. Phys.* 60.8 (2019), pp. 083509, 26. ISSN: 0022-2488,1089-7658. MR: [3995715](#). arXiv: [1901.07479](#) [[math-ph](#)]
- T. Bothner, A. Its, and A. Prokhorov. “On the analysis of incomplete spectra in random matrix theory through an extension of the Jimbo-Miwa-Ueno differential”. In: *Adv. Math.* 345 (2019), pp. 483–551. ISSN: 0001-8708,1090-2082. MR: [3899969](#). arXiv: [1708.06480](#) [[math-ph](#)]
- A. R. Its, O. Lisovyy, and A. Prokhorov. “Monodromy dependence and connection formulae for isomonodromic tau functions”. In: *Duke Math. J.* 167.7 (2018), pp. 1347–1432. ISSN: 0012-7094,1547-7398. MR: [3799701](#). arXiv: [1604.03082](#) [[math-ph](#)]
- A. Its and A. Prokhorov. “Connection problem for the tau-function of the sine-Gordon reduction of Painlevé-III equation via the Riemann-Hilbert approach”. In: *Int. Math. Res. Not. IMRN* 22 (2016), pp. 6856–6883. ISSN: 1073-7928,1687-0247. MR: [3632069](#). arXiv: [1506.07485](#) [[math-ph](#)]
- A. O. Prokhorov and N. D. Filonov. “The Maxwell operator with periodic coefficients in a cylinder”. In: *Algebra i Analiz* 29.6 (2017), pp. 182–196. ISSN: 0234-0852. MR: [3723815](#). arXiv: [1801.10440](#) [[math-ph](#)]

- A. Prokhorov and N. Filonov. “Regularity of electromagnetic fields in convex domains”. In: *J. Math. Sci. (N.Y.)* 210.6 (2015), pp. 793–813. ISSN: 1072-3374,1573-8795. MR: [3407793](#). arXiv: [1501.07081](#) [[math-ph](#)]

SUBMITTED TO JOURNAL

- H. Desiraju, P. Ghosal, and A. Prokhorov. *Proof of Zamolodchikov conjecture for semi-classical conformal blocks on torus*. 2024. arXiv: [2407.05839](#) [[math-ph](#)], submitted to ‘[Communications on Pure and Applied Mathematics](#)’
- A. Its and A. Prokhorov. *On $\beta = 6$ Tracy-Widom distribution and the second Calogero-Painlevé system*. 2020. arXiv: [2010.06733](#) [[nlin.SI](#)], submitted to ‘[Pure and Applied Functional Analysis](#)’

NON-REFEREED PUBLICATIONS AND PREPRINTS

- A. R. Its and A. Prokhorov. “On some Hamiltonian properties of the isomonodromic tau functions”. In: *Rev. Math. Phys.* 30.7 (2018), pp. 1840008, 38. ISSN: 0129-055X,1793-6659. MR: [3833049](#). arXiv: [1803.04212](#) [[math-ph](#)]

THESIS

- A. Prokhorov. *Connection Problem for Painleve Tau Functions*. Thesis (Ph.D.)–Purdue University. ProQuest LLC, Ann Arbor, MI, 2019, p. 112. ISBN: 979-8379-67239-3. MR: [4625528](#)

EDUCATION

2019	PhD, Department of Mathematical Sciences at Indiana University-Purdue University Indianapolis. Dissertation: “Connection problem for Painlevé tau functions”. Advisor: Alexander Its.
2014	Master of Physics, St. Petersburg State University. Thesis: “Regularity of electromagnetic fields in convex domains”. Advisor: Nikolai Filonov.

WORK EXPERIENCE

2025–2026	Visiting Assistant Professor, Department of Mathematical Sciences, University of Cincinnati.
2024–2025	Postdoctoral Scholar, Department of Statistics, University of Chicago.
2021–2024	NSF Postdoctoral Fellow based at the University of Michigan, Ann Arbor.
09/2021–12/2021	Postdoctoral fellow, Mathematical Sciences Research Institute, Berkeley.
09/2019–06/2021	Postdoctoral Assistant Professor, Department of Mathematics, University of Michigan, Ann Arbor.
01/2017–	Researcher, Saint-Petersburg State University.
09/2011–12/2012	Research Assistant at the Chebyshev Laboratory at the Saint-Petersburg State University.

SERVICE

- 12/2024 Organizer of the special session "Discrete and continuous integrable systems: geometry analysis and applications" at the Joint meeting of NZMS, AustMS and AMS, December 9-13, 2024, University of Auckland, New Zealand.
- 06/2022 Organizer of the Summer School on Random Matrices, University of Michigan, Ann Arbor, June 13-24, 2022.
Other organizers: Jinho Baik, Raj Nadakuditi.
- 06/2024 Organizer of the Summer School on Random Matrices, University of Michigan, Ann Arbor, June 17-28, 2024.
Other organizers: Jinho Baik, Raj Nadakuditi.
- 09/2022 Organizer of the conference, "The charm of integrability - Honoring the scientific contributions of Alexander Its on the occasion of his 70th birthday" University of Bristol, UK, September 12-16, 2022.
Other organizers: Tamara Grava, Thomas Bothner, Ken McLaughlin.
- 09/2020–06/2024 Organizer of the seminar on integrable systems and random matrix theory at the University of Michigan.
Other organizers: Ahmad Barhoumi, Guilherme Silva, Jinho Baik, Peter Miller.
- 09/2020–12/2020 Mentor of the undergraduate research project "Unraveling the patterns of Painlevé zeros" in the Laboratory of Geometry.
Other mentors: Jörn Zimmerling, Elizabeth Collins - Woodfin, Benjamin Krakoff.
Students: Hexin Cui, Wenhao Deng, Xiaoqi Peng
- 05/2021–06/2021 Mentor of the REU project "Computing The Constant In The Left-tail Asymptotic Of Maximum Eigenvalue Distribution Of Finite GUE".
Other mentor: Fred Adams.
Student: Xiaoqi Peng.
- 05/2023–06/2023 Mentor of the REU project "Small x asymptotics for special function solutions of Painlevé-III equation".
Student: Hao Pan.
Publication: H. Pan and A. Prokhorov. "Asymptotic properties of special function solutions of the Painlevé III equation for fixed parameters". In: *Stud. Appl. Math.* 154.4 (2025), Paper No. e70051, 32. ISSN: 0022-2526, 1467-9590. MR: 4897093. arXiv: 2407.04852 [math.CA]
- Guest editor of "Special Issue on Evolution Equations, Exactly Solvable Models and Random Matrices in honor of Alexander Its' 70th birthday"
- Referee:
 - Annales Henri Poincaré
 - Communications in Mathematical Physics
 - Nonlinearity
 - Proceedings of the American Mathematical Society
 - SIAM Journal on Mathematical Analysis
 - Letters in Mathematical Physics
- American Mathematical Society Graduate Student Chapter at Indiana University-Purdue University Indianapolis (<https://sites.google.com/iu.edu/amsiupui>)
 - 09/2017–06/2018 President
 - 09/2016–06/2017 Vice-President
 - 09/2015–06/2016 Secretary

TEACHING EXPERIENCE

- 05/2026–08/2026 Teaching STAT 2037 (Probability & Statistics I) at University of Cincinnati.
- 01/2026–05/2026 Teaching STAT 1031 (Introduction to Statistics) at University of Cincinnati.
- 09/2025–12/2025 Teaching Math 1061 (Calculus I) at University of Cincinnati.
- 01/2024–08/2024 Teaching Math 471 (Introduction to Numerical Methods)
at University of Michigan, Ann Arbor.
- 01/2023–05/2023 Teaching Math 354 (Fourier analysis and its applications)
at University of Michigan, Ann Arbor.
- 09/2022–12/2022 Teaching Math 454 (Boundary value problems for partial differential equations)
at University of Michigan, Ann Arbor.
- 01/2019–05/2021 Teaching Math 216 (Introduction to differential equations)
at University of Michigan, Ann Arbor.
- 09/2019–12/2019 Teaching Math 115 (Calculus I)
at University of Michigan, Ann Arbor.
- 01/2019–05/2019 Teaching Math 15400 (Trigonometry)
at Indiana University-Purdue University Indianapolis.
- 09/2018–12/2018 Teaching Math 15300 (College Algebra)
at Indiana University-Purdue University Indianapolis.
- 09/2018–12/2018 Teaching Math 11000 (Fundamentals of Algebra)
at Indiana University-Purdue University Indianapolis.
- 09/2017–05/2018 Teaching Math M118 (Finite Mathematics)
at Indiana University-Purdue University Indianapolis.
- 05/2017–06/2017 Teaching Math 51000 (Vector Calculus)
at Indiana University-Purdue University Indianapolis.
- 08/2016–12/2016 Teaching Math 17100 (Multidimensional Mathematics)
at Indiana University-Purdue University Indianapolis.

HONORS/AWARDS

- First Year Fellowship from School of Science, IUPUI, 2014.
- Outstanding Advanced Mathematics Graduate Student, IUPUI, 2016.
- Charalambos D. Aliprantis Prize, IUPUI, 2017. (This scholarship is awarded to mathematics graduate students who exemplify outstanding scholastic achievements as well as leadership qualities.)
- Yuri Abramovich Memorial Scholarship, IUPUI, 2018. (This scholarship supports continuing undergraduate and graduate students who have a keen interest in the study of mathematics, who demonstrate academic excellence, especially in mathematics courses beyond the sophomore level and who show promise for a career in mathematics.)
- Outstanding Advanced Mathematics Graduate Student, IUPUI, 2019.
- [NSF Postdoctoral Fellowship](#), 2021-2024

RESEARCH PRESENTATIONS

- [The Kardar-Parisi-Zhang Universality Class & Related Topics](#), workshop at Brin Mathematics Research Center, April 20-24, 2026. Talk: "Left Tail Asymptotics of the Two-Point Distribution of the Airy Process"
- Baylor University, Departmental Colloquium, February 4, 2026. Talk: "Painlevé equations and their generalizations"
- University of Michigan, Integrable systems and random matrix seminar, December 8, 2025. Talk: ["Connection formulas for the Heun equation from Painlevé theory"](#)
- AMS Fall Southeastern Sectional Meeting, Tulane University, New Orleans, LA, October 3-5, 2025. Talk: ["Asymptotic analysis of orthogonal polynomials inspired by the matrix model for JT gravity"](#)
- University of South Florida, Departmental Colloquium, March 7, 2025. Talk: ["Nonlinear steepest descent on a torus: A case study of the Landau-Lifshitz equation"](#)
- University of Utah, Stochastics seminar, February 28, 2025. Talk: ["Proof of Zamolodchikov conjecture for semi-classical conformal blocks on the torus"](#)
- University of Chicago, Probability and Statistical Physics Seminar, February 21, 2025. Talk: ["Proof of Zamolodchikov conjecture for semi-classical conformal blocks on the torus."](#)
- SIAM Conference, Texas-Louisiana Section, 7th Annual Meeting, October 11-13, 2024, Baylor University. Talk: ["Low temperature asymptotic of partition function for logarithmic gas on the torus"](#)
- Talk at the IUPUI AMS student chapter ["Semiclassical analysis of conformal blocks on the torus."](#), May 17, 2024.
- AMS Spring Central Sectional Meeting, University of Cincinnati, Cincinnati, OH, April 15-16, 2023. Talk: ["Large time asymptotic for solutions of Landau-Lifshitz equation using Riemann-Hilbert approach"](#)
- AMS Joint Mathematics Meeting, Boston, January 4-7, 2023. Talk: ["Asymptotical properties of rational solutions of Painlevé-III \(\$D_6\$ \) equation and application to modulated bi-orthogonal polynomials."](#)
- [Midwestern Workshop on Asymptotic Analysis](#), Purdue University Fort Wayne, October 7-9, 2022. Talk: "Monodromy Map under the Confluence $\text{PIII}(D_6) \rightarrow \text{PIII}(D_8)$ ".
- [The Twelfth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory](#), University of Georgia, Athens, GA, USA, March 30 - April 1st, 2022. Talk: "Large parameter asymptotic of rational solutions of Painlevé III (D_6) equation near zero".
- AMS Spring Central Virtual Sectional Meeting, March 26-27, 2022. Talk: ["Integrable systems governing KPZ fixed points"](#).
- Michigan State University, Mathematical Physics seminar, November 9th, 2021. Talk: "On $\beta = 6$ Tracy-Widom distribution and the second Calogero-Painlevé system"
- Mathematical Sciences Research Institute Seminar, December 3rd, 2021. Talk: ["Integrable structure for the Multitime distribution of TASEP"](#)
- Mathematical Sciences Research Institute Mini Course, September 2nd, 2021. Talk: ["Riemann-Hilbert problems application in the random matrix theory"](#)

- [Asymptotic methods in Mathematical Physics](#), Conference dedicated to the memory of V. S. Buslaev, EIMI, Saint-Petersburg, June 20th - 22nd, 2021
Talk: “ Integrable structure for the multipoint distribution of TASEP”.
- [Integrable systems in Geometry and Mathematical Physics](#), Conference in memory of Boris Dubrovin, SISSA, Trieste, June 28th - July 2nd, 2021
Virtual 3 minute talk: “ Large parameter asymptotics of rational solutions of Painlevé III equation near zero”.
- IU Analysis seminar, Bloomington, March 17th, 2021
Virtual talk: “ Behavior of rational solutions of Painlevé III equation near zero”.
- [Bernoulli-IMS One World Symposium](#), Virtually, August 24th - 28th, 2020.
Talk: “ On $\beta = 6$ Tracy-Widom distribution and the second Calogero-Painlevé system. ”.
- [Junior Integrable Probability Seminar](#), Virtually, July 9th, 2020.
Talk: “ Integrable structure behind the multitime KPZ fixed point distribution. ”.
- Workshop “[Complex analysis in mathematical physics and applications](#)”, Isaac Newton Institute for Mathematical Studies, Cambridge, UK, October 28th - November 1st, 2019.
Poster: “Asymptotic of solution of three-component Painlevé II equation”.
- Forty-Seventh Annual Mathematics Conference “Differential Equations and Dynamical Systems and their Applications”, Miami University, Oxford, OH, USA, September 20 - 21, 2019.
Talk: “Connection problem for Painlevé tau functions.”.
- Workshop “[Painlevé equations in the Midwest](#)”, University of Michigan, Ann Arbor, MI, USA, August 23 - 24, 2019.
Talk: “Asymptotic of solution of three-component Painlevé II equation”.
- [The Eleventh IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory](#), University of Georgia, Athens, GA, USA, April 17 - 19, 2019.
Talk: “Asymptotic of 3-component Painlevé-II equation”.
- [AMS Fall Central Sectional Meeting](#) , University of Michigan, Ann Arbor, October 20-21, 2018.
Talk: “On some Hamiltonian properties of isomonodromic tau functions”.
- [Midwestern Workshop on Asymptotic Analysis](#), IU, Bloomington, October 5-7, 2018.
Poster: “On some Hamiltonian properties of isomonodromic tau functions”.
- Workshop “[Tau Functions of Integrable Systems and Their Applications](#)”, BIRS, Banff, Canada, September 2-7, 2018.
Talk: “On some Hamiltonian properties of isomonodromic tau functions”.
- Invited speaker at the probability seminar at University of Virginia, Charlottesville, October 25, 2017.
Talk: “Limiting distribution of smallest eigenvalue of thinned complex Wishart matrices”
- [Midwestern Workshop on Asymptotic Analysis](#), IUPUI, Indianapolis, October 6-8, 2017.
Poster: “The smallest eigenvalue distribution of incomplete Laguerre Unitary Ensemble” .
- [School on Dyson-Schwinger equations, topological expansions, and random matrices](#), Columbia University, New York, August 28 - September 1, 2017.
Poster: “The smallest eigenvalue distribution of incomplete Laguerre Unitary Ensemble” .
- Graduate Summer School on Random Matrices at PCMI, Utah, Park city, June 25 - July 15, 2017.
Poster: “The smallest eigenvalue distribution of incomplete Laguerre Unitary Ensemble” .

- School on “[Quantum integrable systems, conformal field theories and stochastic processes](#)”, Institut d’Études Scientifiques de Cargèse, Cargèse, France, September 12-23, 2016.
Talk “Asymptotics of tau-function for Painlevé equations”.
- Workshop “Moduli spaces, integrable systems, and topological recursions”, CRM, Montréal, Canada, January 9-13, 2016.
Talk “Connection problem for the isomonodromic tau-function of the Sine-Gordon reduction of Painlevé-III equation”.
- Workshop “Asymptotics in integrable systems, random matrices and random processes and universality”.
In honour of Percy Deift’s 70th birthday.
CRM, Montréal, Canada, June 7-11, 2015.
Poster “Connection problem for the tau-function of the Sine-Gordon reduction of Painlevé-III equation via the Riemann-Hilbert approach”.
- [6th St. Petersburg Conference in Spectral Theory](#),
dedicated to the memory of M. Sh. Birman.
Russia, St. Petersburg, July, 3-8, 2014.
Talk “Regularity of electromagnetic fields in nonsmooth domains”.
- [Crimean International Mathematical Conference](#).
Ukraine, Crimea, Sudak, September 22 - October 4, 2013.
Talk “Regularity of electromagnetic fields in nonsmooth domains”.
- Annual International Conference “[Days on Diffraction](#)”.
Russia, St. Petersburg, May, 27-31, 2013.
Talk “On absolute continuity of spectrum of the periodic Maxwell operator in a cylinder.”
- [The Twenty Third Crimean Autumn Mathematical School-Symposium](#).
Ukraine, Crimea, Laspi-Batiliman, September, 17-29, 2012.
Talk “The Maxwell operator in the waveguide with periodic coefficients”.